ASPECT WAVE LLC

Vessel Measurement Recorder

November 2026 · Prepared by Sophie Scopazzi · Firmware v8.12

System Overview

The VMR is an open source, small, battery-powered, weather-resistant instrument that records how a vessel moves, where it is, basic information about the weather and ocean waves/swells. It lets users mark the exact time and place that something happens with one button press. It builds its own Wi-Fi network; each connected device becomes a per-user configurable live dashboard. The dashboard also allows data plotting and download. No internet connection is required. There is a display on the unit and physical buttons for non-Wi-Fi data viewing and control.

Each VMR is a general-purpose measurement recorder that has been configured for use on a vessel. The same base with different sensors works as a deployment logger or as a handheld, backpack or vehicle-mounted location and/or weather station logger for land-based deployments.

Marking Locations (and associated data): A problem it solves

On research cruises, teaching boat-based classes, and small-boat fieldwork, positions and times are often still recorded by hand: someone uses a clipboard and reads the ship's GPS display or takes photos of a handheld GPS unit to sort through and digitize later. The paper gets wet or lost, the numbers have to be manually typed in afterwards, and when the team is busy or short-handed the record can be cumbersome to accomplish in a timely manner.

A continuous logger alone does not solve these problems. It would capture everything, but afterwards someone would still have to work backwards from a physical log or from memory when each event or cast happened.

The VMR does both at once. It logs continuously and records marked events in separate files, so the full record and the mark record of what happened (where, when, and other data) is a ready-made CSV file the moment the button is pressed, already cross-referenced to the continuous data with shared annotations.

Capabilities: Motion and stability

- **Live roll and pitch** from a BNO085 orientation sensor, with roll period, significant roll and largest roll in graph (with controllable axis limits) and text format
- **Clinometer** with drag pointers that hold the largest roll each way, on the OLED and on the dashboard. The scale can be zoomed in down to ±10°
- **Acceleration and G-meter:** linear acceleration (vertical, fore-and-aft, athwartships) is logged, and a circular G-meter shows the forces felt by the VMR
- **Stability:** metacentric height (GM) estimated from the roll period and the ship's particulars, the wave encounter period, heave, and a check for synchronous or parametric rolling
- **Waves:** The wave encounter period and estimated wave/swell height (Hs)
- **Dock mode:** trim in feet and list in degrees while alongside the dock, calibrated against the vessel's draft marks
- **Yaw and rate of turn** are relative (the magnetometer is not fully implemented yet)

Position and location marks

- **GPS:** position, speed and course over ground, altitude, fix status, and a satellite sky plot. Coordinates can be shown as decimal degrees, degrees and decimal minutes, or degrees, minutes and seconds
- **One-press marks:** a physical button on the box, or a button on the Wi-Fi dashboard, records the time, position, and other data at the time (weather, etc.) to a separate marks CSV file
- **Start and stop for casts:** a second button records the start then stop of a system deployment as one event in one row, with the start and end positions, the duration, and the drift distance between them automatically calculated and added to the mark CSV
- **Personal Annotations:** fillable notes field on each mark, for example the tree species, the station name or the instrument, added via the Wi-Fi dashboard
- **Position quality:** satellites, HDOP, and age of fix saved with every mark
- **Linked to the continuous log:** while continuously recording, each mark is also written into the continuous file, so the two line up without any matching by hand
- **Delete last mark** with a confirmation step, for an accidental press. The marks are saved to a CSV file after each addition or deletion, so marks survive a restart of the VMR

Weather

- Barometric pressure, air temperature, relative humidity and dew point, from I2C add-on sensors (BMP388 and SHT30). These are optional; the VMR works with or without them

Recording and data access

- **Continuous logging** of everything to the microSD card at 10 Hz, as CSV, with times in UTC from a GPS-disciplined real-time clock. GPS fields are filled only on rows with a new fix, so position data is not duplicated. The 10Hz logging is for the BNO085 sensor; without, the VMR can be configured with slower logging periods via the Wi-Fi dashboard
- **Self-describing files:** header lines record the firmware build and other metadata as comments, so the files load directly into Python (pandas) or a spreadsheet
- **Plots on the device:** recorded files can be plotted by individual variable from the Wi-Fi dashboard without first downloading anything. The plots can be saved to the VMR or downloaded to any connected device via the Wi-Fi dashboard. Plots can be created while the VMR is logging and of the currently-open record file
- **Wireless download of data files:** all files (recordings, plots, mark files) are downloadable over the VMR's Wi-Fi dashboard
- **Persistent Storage:** Maxima and roll statistics, vessel particulars, settings for plots, and other information is stored in EEPROM for continuity between VMR restarts and firmware updates

Interfaces

- **Wi-Fi dashboard:** works on any phone, tablet or laptop browser, with a day, night, and red mode. A welcome screen explains the main features the first time a device connects
- **OLED screen and buttons:** the VMR is fully usable without a phone. One physical button rotates through OLED screens that show live readings from sensors, the roll graph, position, maxima, the clinometer, the G-meter, and the location marks. A physical knob turns the screen on and off, and controls the brightness of the OLED and indicator LEDs

- **Clock:** set from GPS, or from the connected phone or laptop until GPS time is available. The local time zone is automatically detected when a device connects to the Wi-Fi dashboard, and is changeable via the same
- **Firmware updates** via Wi-Fi or USB-C connector

Power and operating modes

- **Battery powered**: charged via USB-C. Each VMR has a built-in battery gauge with charge or discharge rate shown on the Wi-Fi dashboard. The VMR can be plugged-in continuously
- **Standard Mode**: following a restart, the system powers into live-data viewing without data logging and with the Wi-Fi at full power
- **Datalogger Mode:** for long unattended recordings, the Wi-Fi dashboard and the screen switch off, and the processor slows down while recording continues. The Mark button returns the system to Standard mode
- **Charge Mode:** everything powers down so the battery charges as fast as possible. The only indication the battery is fully charged is an LED indicator light turning off

One platform, several configurations

- The Wi-Fi dashboard is built from cards driven by JSON data. A card can be put together for any new sensor. The hardware works the same way: the same core hardware (processor, GPS, clock, storage, screen, Wi-Fi) uses different sensors for different applications.

Configuration	Sensors	Main job
Vessel Measurement Recorder (full)	Core + orientation sensor + weather sensors	Permanent bridge instrument: clinometer, motion, stability, weather and continuous record
VMR without weather	Core + orientation sensor	Motion and stability only (cheaper)
VMR inclinometer only	Core - GPS, clock, storage + orientation sensor	Inclinometer (live data only with no storage, location, or time)
Weather Station	Core + weather sensors	Live weather and permanent record
Deployment Logger	Core only	Marks and casts for science deployments, on the water or on land
Custom builds	Core + any I2C or UART sensor	Other measurements, with respective dashboard cards

Audiences and use cases

Audience	Use case	Key features
Deck officers, bridge teams, and vessel operators	Permanently mounted on the vessel as a clinometer and motion recorder, with weather data alongside. Gives a record of roll and sea state over a voyage, a check on stability while under way, and trim and list at the dock	Clinometer, roll period, GM estimate, synchronous roll check, dock mode, night mode, datalogger mode for long passages, max roll experienced (all time, x min window)
Science parties on research vessels and citizen scientists	Marking both individual events and when an instrument goes over the side and comes back. CTD and other system casts	Mark button, cast start and stop, notes, continuous record and separate marks CSV linked to the cont. log via shared annotations
Maritime academies and classes	To record stations and deployments quickly and accurately. Teaching tool for ship motion and stability	Phone dashboard for several users, marks, live graphs
Field oceanographers on small boats with small teams	Fieldwork where there are few hands available. One button press records the start and end of each cast, with no paper, typing or photos of a handheld GPS	Physical buttons, cast duration and drift, battery power, no phone required
Terrestrial field scientists	Carried by hand or attached to a backpack to mark the location of every tree, plot or sample using a physical button or phone	Marks with notes, coordinate format choice, Deployment Logger configuration
Instrument builders and open-science users	An open source and low-cost base for adding other sensors and dashboard cards, built from parts that are easy to buy	Off-the-shelf modules, I2C add-ons, UART expandability, JSON-driven dashboard cards

Controls on the VMR

Control	Action
Mark / Cast buttons	A mark; hold Cast and press Mark to start or stop a cast. Mark leaves Datalogger mode
Screen button or OLED button B	Next screen. B resumes a paused VMR
Pot and pot switch	OLED and LED brightness; OLED on / off (off also turns the LEDs off)
A / C on screen 2	Roll graph top up / bottom down, 5° steps
Hold A + C for 3 s on screen 3	Level zero on / off
Hold A + C for 3 s on screen 4	Reset the all-time maxima
Hold A + C for 3 s on screen 6	Reset the clinometer drag pointers
Hold A + C for 3 s on screen 10	Reset the G-meter peaks

Screens 4, 8 and 9 are picked from the dashboard (card 3); the screen button skips them.

Indicator LEDs

LED	Shows
Mark	One blink at a mark. On during a cast, with a short off-blink for a mark made during casts. A short flash when the Mark button wakes the VMR from Datalogger mode.
Recording	On while recording. Slow blink when paused, fast blink when the microSD card is failing, a blink per row in Datalogger sleep.
Fault	FAULT in Morse (..-. .- ..- .-.. -) while something has been wrong for 5 s: the IMU, GPS module or an add-on silent; the OLED, EEPROM or RTC missing; the microSD card failing while recording; the battery under 20 % and not charging. It finishes the word once all is well.

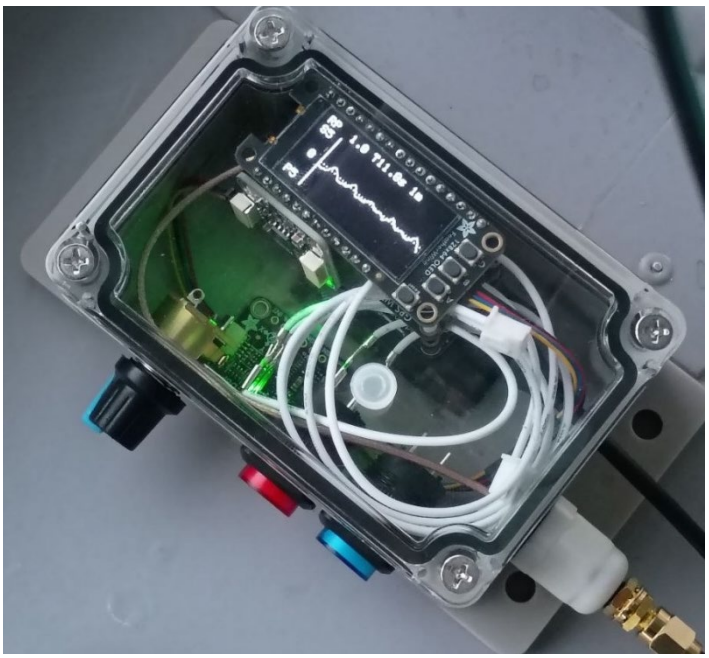
Brightness follows the OLED (the knob, or the dashboard slider) and never goes fully dark. The pot switch turns all LEDs off; they are also off in Datalogger sleep and Charge Mode.

Software

The VMR is built in C++ and JSON and uses the Arduino IDE to compile the firmware. The firmware is a single Arduino sketch for the ESP32-S2 that uses Adafruit's open-source libraries for each module. The dashboard is a web page (HTML, CSS and JavaScript) stored inside the firmware and served by the VMR itself. The page and the device exchange live readings and settings as JSON. The sensor and logging loop and the web server run as separate tasks on the processor.

System Photos







Bill of Materials

No custom circuit boards are required. Every part connects either by stacking Feather boards on top of each other or a shared I2C cable chain (STEMMA QT / Qwiic).

Function	Part	Num	Cost
Processor and Wi-Fi	Adafruit ESP32-S2 Feather w/ w.FL to SMA adapter and external Wi-Fi antenna	6282, 5444, 945	$29.5
Position	Ultimate GPS FeatherWing (PA1616D) w/ external active antenna	3133, 960	$47
Clock and storage	Adalogger FeatherWing (precision real-time clock w/ CR1220 and microSD)	2922, 380, 5252 / 1294	$15 / $33.5
Display	OLED FeatherWing (128 × 64) with buttons + reset button	4650	$15
Motion	BNO085 9-axis orientation sensor (I2C)	4754	$29.5
Persistent memory	24LC32 EEPROM (I2C)	5146	$4
Weather – Pressure	BMP388 pressure (I2C),	3966	$10
Temperature & humidity	SHT30 temperature and humidity connected w/ Qwiic / STEMMA QT breakout board (I2C)	5064, 5961	$11
GPIO Expansion Board	PCF8574 GPIO for mark and cast buttons (and extra ports)	5545	$5
ADC Expansion Board	ADS7830 ADC for dimmer pot	5836	$6
OLED + function LEDs power / dimmer	10k linear potentiometer with built-in switch	3395	$2
System power switch	Latching power switch	1476	$2
Screen & Mark button	Momentary pushbuttons (2x)	1439	$3
LED indicator lights	4 panel-mount 3.3 V LEDs, multicolor pack (red / green used)	Many vendors	$1
Wires	STEMMA QT 50 mm (x6) and arcade quick-connect wires	4399, 1152	$11
Terminal Blocks	Terminal blocks, 2.54 mm pitch: one 4-pin, two 8-pin	2137, 2141	$7
Power	2,500 mAh – 10,000 mAh LiPo battery with gauge w/ USB-C small round panel-mount extension cable	328 or 5035, 6069	$19.5 / $34.5
Housing	Weatherproof enclosure	Pelican 1040	$28
Miscellaneous parts for affixing and etc.	Double-sided Velcro, black nail polish (optional, to shade green always-on power LEDs on I2C boards)	Many vendors	$10
	(full) **Vessel Measurement Recorder**	**$256 / $290**	
	(without weather) **Vessel Measurement Recorder**	**$235 / $269**	
	(inclinometer) **Vessel Motion Recorder**	**$173 / $188**	
	(no IMU, weather sensors only) **Weather Station**	**$203 / $239**	
	(no sensors except GPS) **Deployment Logger**	**$145 / $181**	

Vessel Measurement Recorder

Dashboard Cards & Build Guide

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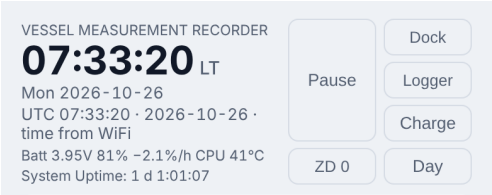
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Wi-Fi Dashboard

There dashboard is broken into cards. Every card can be hidden / shown as desired using a permanent card at the bottom of the dashboard. The information at the top outside of the cards cannot be hidden

Top Information and Buttons



The local time using the ZD retrieved from the connected device or manually set by tapping the button. Buttons for dock, logger, and charge mode and to change the display color (day, night, red)

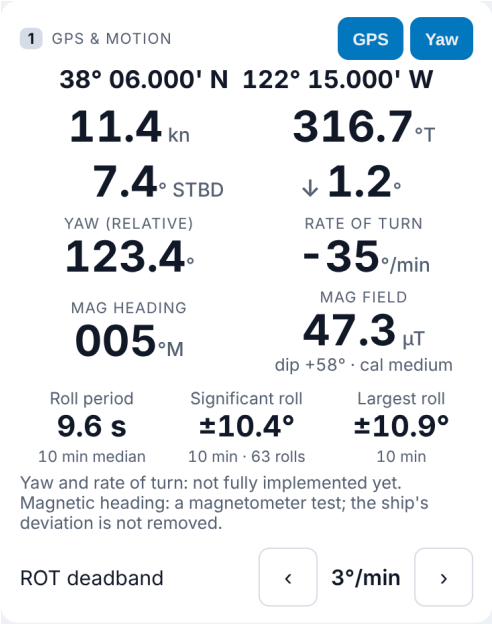
Dock Mode shows a hidden trim and list card at the top (viewing this card is only accessible via the button)

Logger Mode disables everything not required to save data to the microSD card (Wi-Fi, with longer sample-durations the GPS, and more)

Charge Mode turns everything off for fastest charging

Pause Button pauses the current recording, in case the VMR needs to be moved while recording. It does not pause the live data feed shown on the dashboard

Card 1 – GPS & Motion



Show and hide the GPS and Yaw / Magnetic data with buttons top right

With Yaw shown, the card also shows the magnetic heading and the magnetic field from the BNO085's magnetometer, which are logged in the CSV. The ship's own deviation is not removed

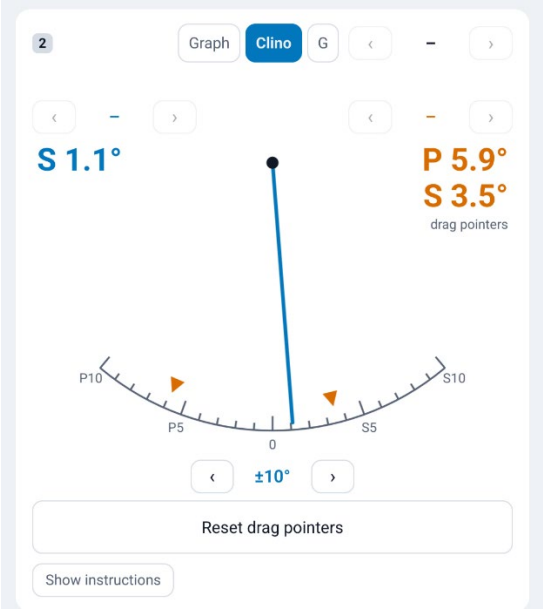
Variation is shown and input with a button between the magnetic heading and magnetic field (not yet in photo to left)

Card 2 – Live Data Graphs

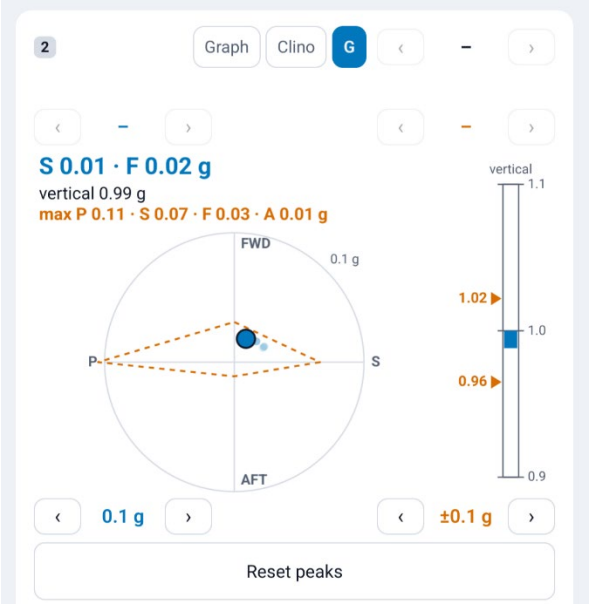


Graph

The OLED graph, clinometer, and g-meter mirrors the dashboard settings

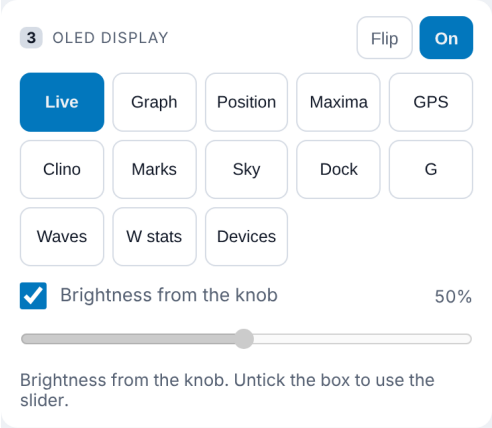


Clinometer



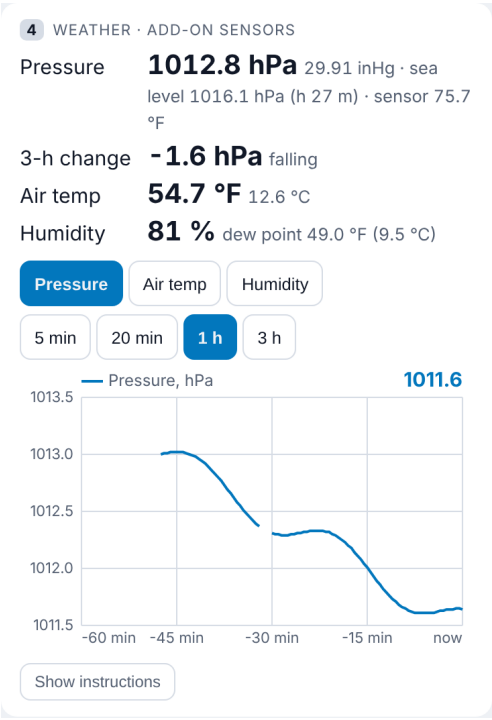
G-meter

Card 3 – OLED Display



Power on and off and 180° flip the OLED. Select what view to see, with brightness controlled from the Wi-Fi dashboard or from the knob

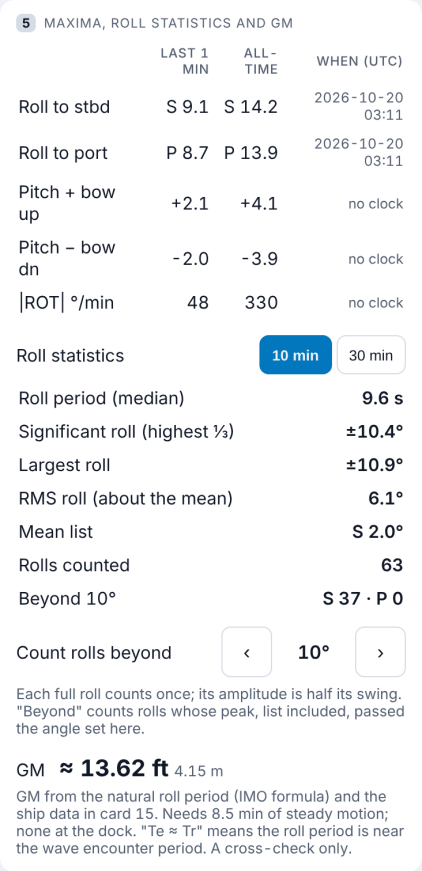
Card 4 – Weather



Pressure (also in inHg and reduced to sea level, if the vessel particulars, drafts, and VMR location onboard are entered), the 3-hour change, air temperature, humidity and dew point from the BMP388 and SHT30. Shown only when one of those sensors is switched on in card 17.

Tap Pressure, Air temp or Humidity for a live graph over 5 or 20 minutes, 1 or 3 hours. The dew point is drawn dashed with the air temperature: where they meet, expect fog or condensation.

Card 5 – Maxima, Roll Statistics and GM



5 MAXIMA, ROLL STATISTICS AND GM

	LAST 1 MIN	ALL-TIME	WHEN (UTC)
Roll to stbd	S 9.1	S 14.2	2026-10-20 03:11
Roll to port	P 8.7	P 13.9	2026-10-20 03:11
Pitch + bow up	+2.1	+4.1	no clock
Pitch − bow dn	-2.0	-3.9	no clock
|ROT| °/min	48	330	no clock

Roll statistics [10 min] [30 min]

Roll period (median)	9.6 s
Significant roll (highest ⅓)	±10.4°
Largest roll	±10.9°
RMS roll (about the mean)	6.1°
Mean list	S 2.0°
Rolls counted	63
Beyond 10°	S 37 · P 0

Count rolls beyond ‹ 10° ›

Each full roll counts once; its amplitude is half its swing. "Beyond" counts rolls whose peak, list included, passed the angle set here.

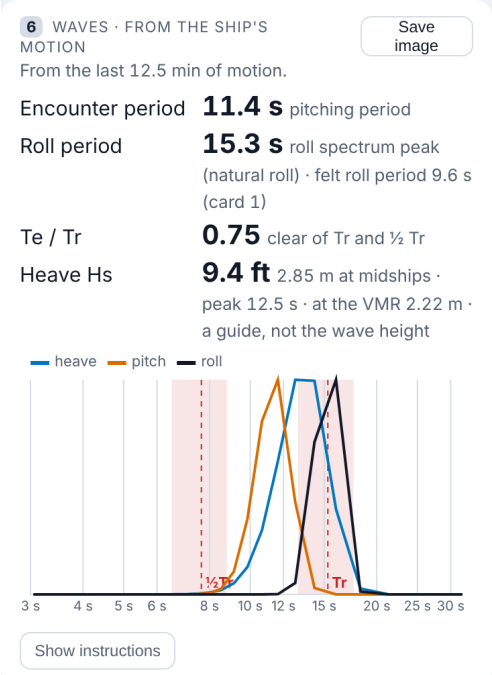
GM ≈ 13.62 ft 4.15 m

GM from the natural roll period (IMO formula) and the ship data in card 15. Needs 8.5 min of steady motion; none at the dock. "Te ≈ Tr" means the roll period is near the wave encounter period. A cross-check only.

Statistics about roll and pitch with ability to count the number of rolls beyond a certain number. Shows the estimated GM. Requires the vessel particulars to calculate

Estimated GM uses

Card 6 – Waves



6 WAVES · FROM THE SHIP'S MOTION [Save image]

From the last 12.5 min of motion.

Encounter period	11.4 s pitching period
Roll period	15.3 s roll spectrum peak (natural roll) · felt roll period 9.6 s (card 1)
Te / Tr	0.75 clear of Tr and ½ Tr
Heave Hs	9.4 ft 2.85 m at midships · peak 12.5 s · at the VMR 2.22 m · a guide, not the wave height

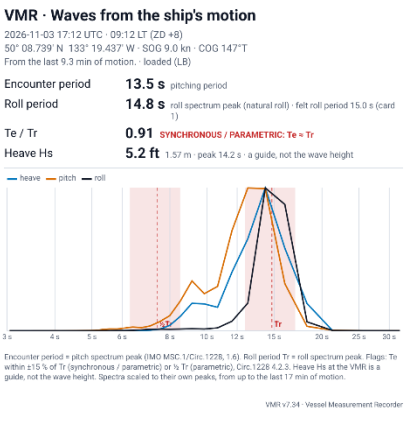
heave pitch roll

Show instructions

Waves graph detailing the encounter period, roll period, Te/Tr (metric for synchronous and parametric rolling) with an estimate of wave height from the heave (Hs)

The wave graph can be saved using the button on the top right

With where the VMR is entered in card 15, Hs is the heave at midships; without it, the heave at the VMR.



VMR · Waves from the ship's motion

2026-11-03 17:12 UTC · 09:12 LT (ZD +8)
50° 08.739' N 133° 19.437' W · SOG 9.0 kn · COG 147°T
From the last 9.3 min of motion. · loaded (LB)

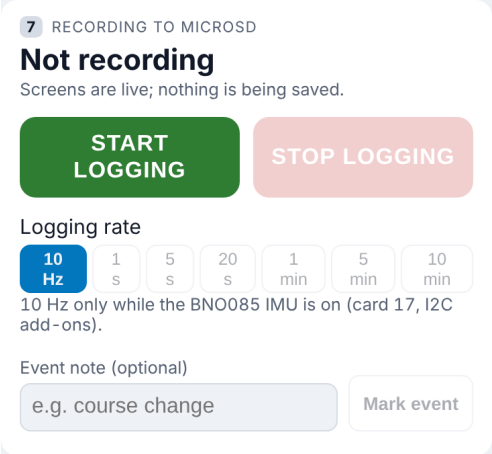
Encounter period	13.5 s pitching period
Roll period	14.8 s roll spectrum peak (natural roll) · felt roll period 15.0 s (card 1)
Te / Tr	0.91 SYNCHRONOUS / PARAMETRIC: Te ≈ Tr
Heave Hs	5.2 ft 1.57 m · peak 14.2 s · a guide, not the wave height

heave pitch roll

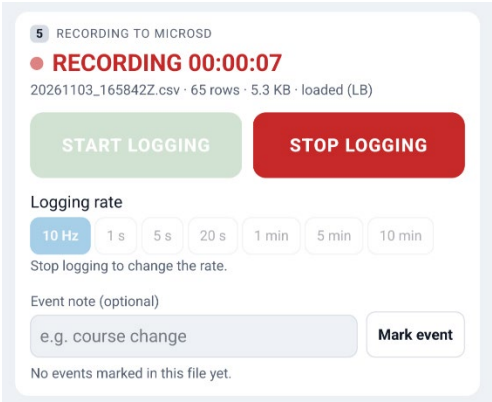
Encounter period = pitch spectrum peak (IMO MSC.1/Circ.1228, 1.6). Roll period Tr = roll spectrum peak. Flags: Te within ±15 % of Tr (synchronous / parametric) or ½ Tr (parametric), Circ.1228 4.2.3. Heave Hs at the VMR is a guide, not the wave height. Spectra scaled to their own peaks, from up to the last 17 min of motion.

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Card 7 – Recording to MicroSD



Not recording



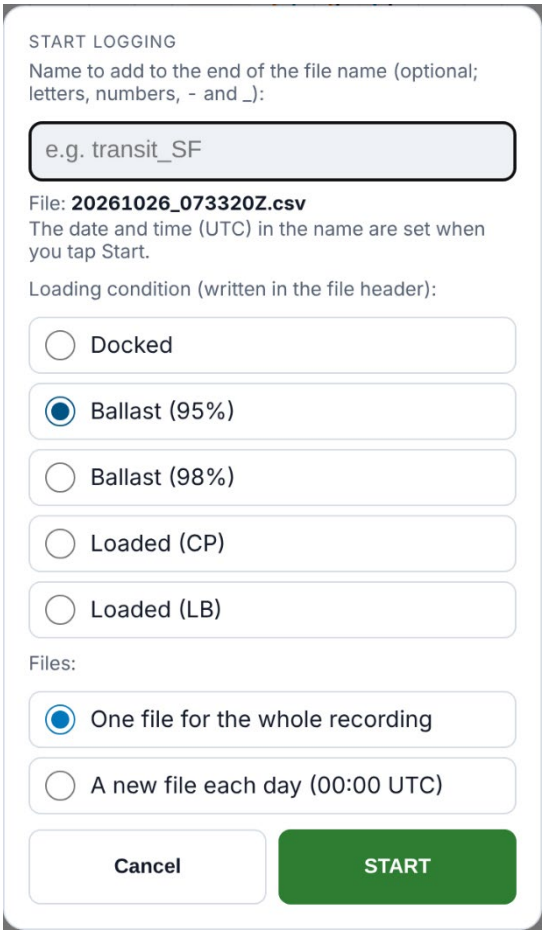
Recording

Start and stop logging. Change the logging frequency. With the IMU enabled the logging rate cannot be less than 10 Hz (other settings grayed out)

When the VMR is continuously logging, an event note can be added in this card that is saved in the CSV

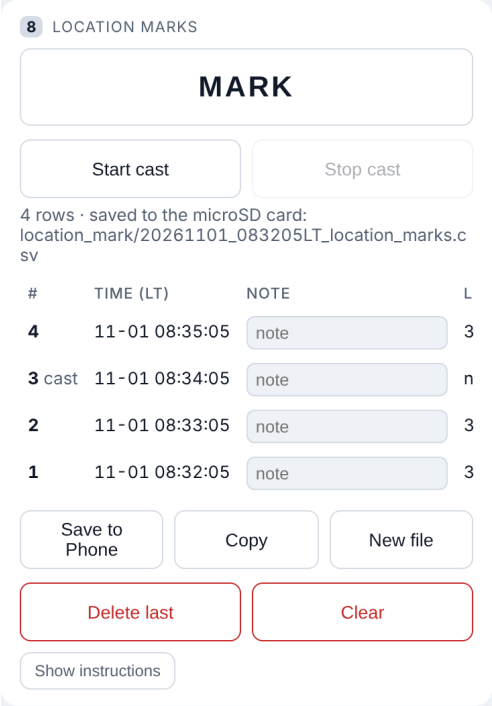
After hitting the green button, a popup appears where an addition can be made to the file name with settings for the ship's load condition (saved in the header) and whether the user desires one file or a new one to be created on each new UTC day.

Once recording, the card says the file name, how many rows it contains, and its size



Start Logging pop-up

Card 8 – Location Marks

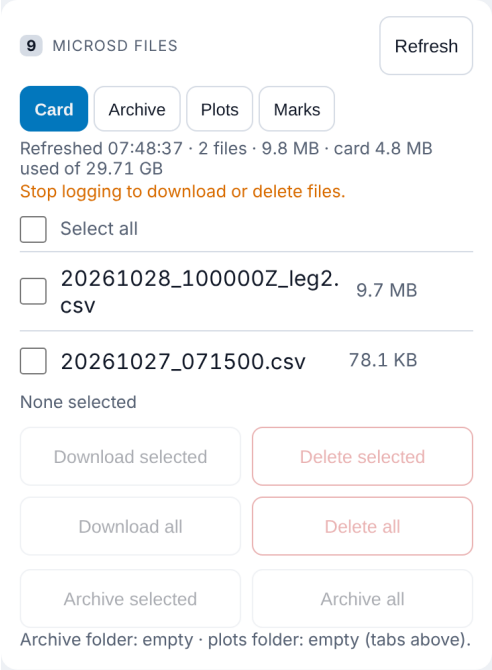


MARK records the time, position, COG / SOG and weather. Start cast and Stop cast record a cast as one row, with its duration and drift are automatically calculated. The physical Mark and Cast buttons on the VMR do the same.

A note can be typed on any row. Save to Phone downloads the marks CSV, Copy puts it on the clipboard, New file starts a new list and keeps the old file, and Clear deletes both. Delete last removes the newest row.

Opened with the Show Location Marks button.

Card 9 – MicroSD Files



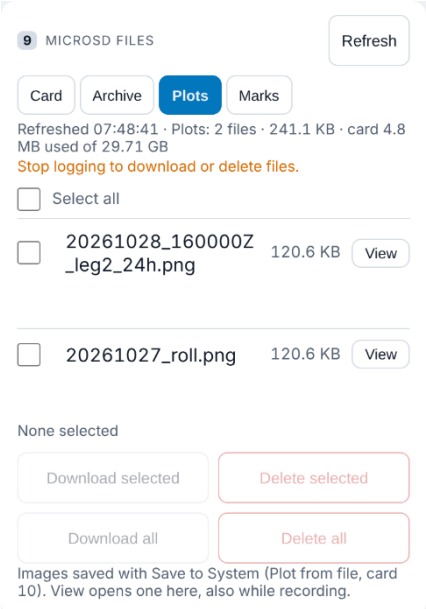
Main card

Viewer for the files on the microSD card. Files can be selected, downloaded, or archived

There is the main card, followed by folders for the archive, plots, and mark files

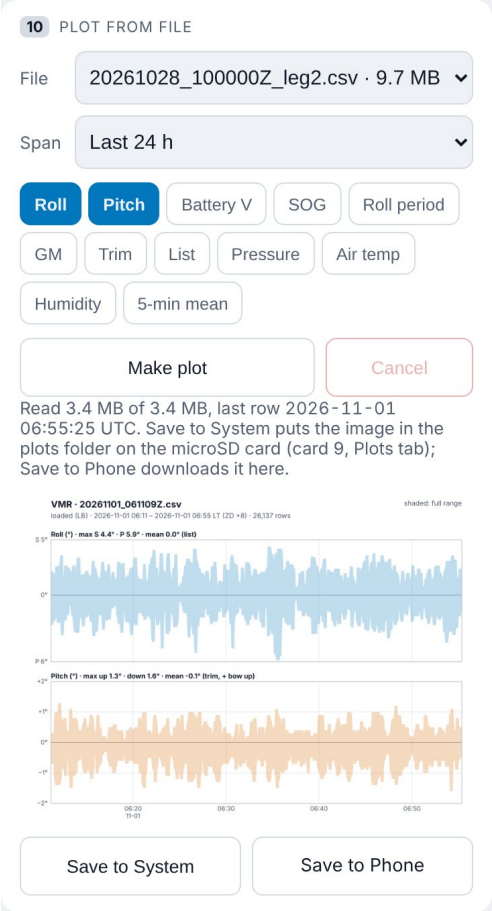
When the VMR is logging, files cannot be downloaded or deleted

On the plots page, images can be viewed by tapping the View button



Plots folder

Card 10 – Plot From File



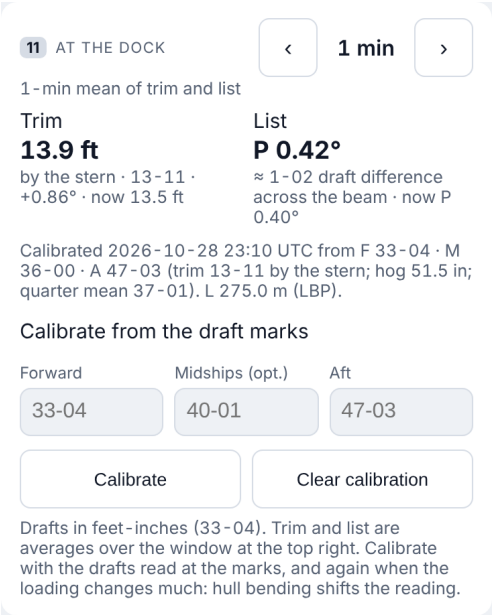
Generate plots from CSV files on the main Card (files not archived)

Select a duration of the file to plot – last 24 hrs, the full file, etc

Once a plot is generated, the user can select the desired variables which are automatically added / removed from the plot in the viewer

Save to System saves the plot to the VMR's Plots folder. Save to Phone saves the plot with the proper file name to the connected device

Card 11 – At the Dock



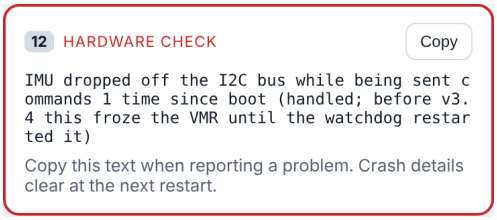
Shown at the top, in place of card 1, while Dock Mode is on (the Dock button). Trim in feet and list in degrees, averaged over 30 s to 5 min (set at the top right).

Calibrate by entering the forward, midships and aft drafts read at the marks. Recalibrate when the loading changes much (hull bending changes the deck slope near the ends). The wave analysis is off while at the dock.

Developer Settings

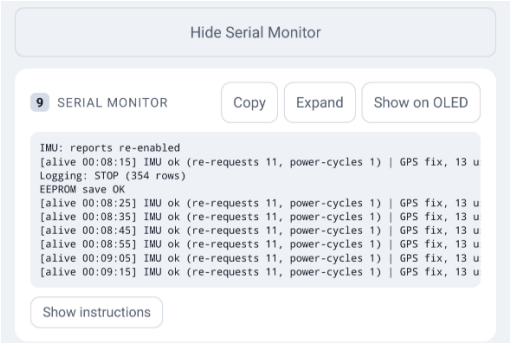
Cards beyond number eleven are hidden behind a Developer Settings button

Card 12 – Hardware Check



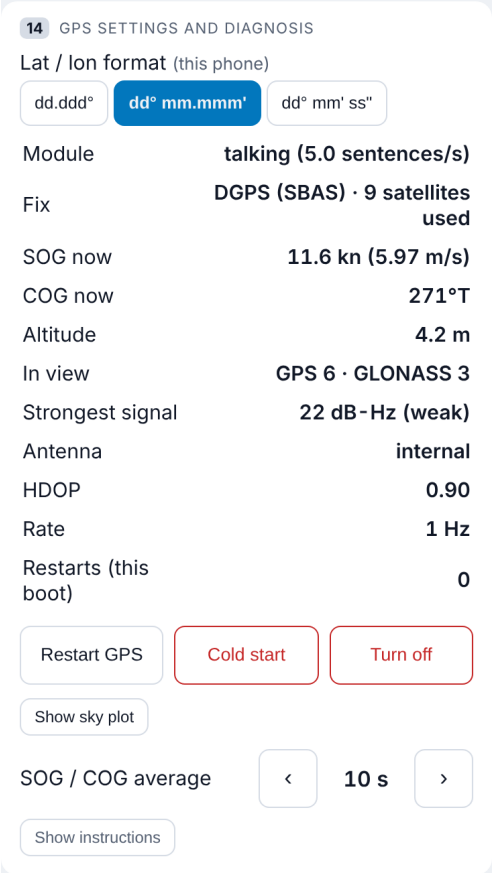
Displays errors with the system. The IMU dropping off the I2C bus is a fairly standard error – the VMR has routines to auto-retry connection

Card 13 – Serial Monitor



By default hidden; only requests data once the card is opened to save bandwidth. Expand makes the card 3x taller, scrolls right with no text wrapping, with Copy button to copy all that is displayed to the connected device's clipboard

Card 14 – GPS Settings and Diagnosis

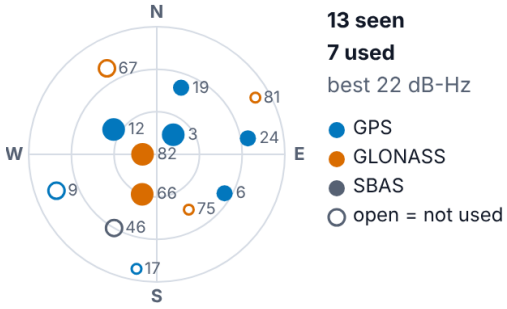


Select the GPS data format that is used to display throughout the VMR's screens. Default format is shown left in image

Health data from the GPS, whether it's talking, etc

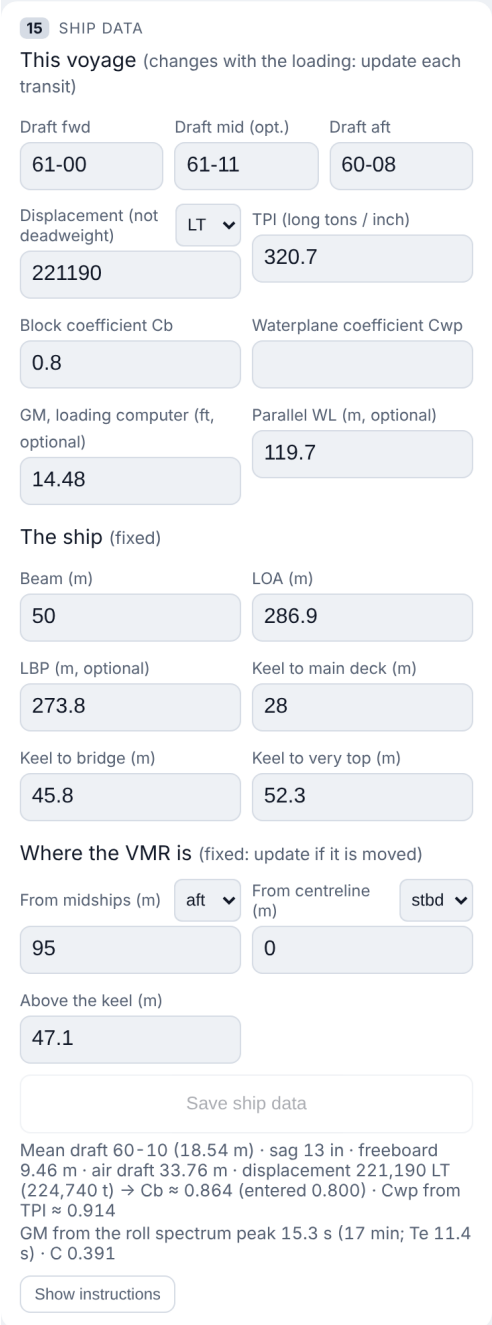
Buttons to restart, cold start, or turn off the GPS

At the bottom, the time to average the SOG / COG over. It defaults to 10 seconds



Satellites the GPS can see: horizon at the edge, overhead in the middle, north up. Filled = used in the fix; bigger = stronger signal.

Card 15 – Ship Data



Per-voyage data at the top and ship-specific data at the bottom (static). All saved to EEPROM to survive system restarts / firmware reflashes

Some values are optional. Without non-optional values input, GM will not be calculated

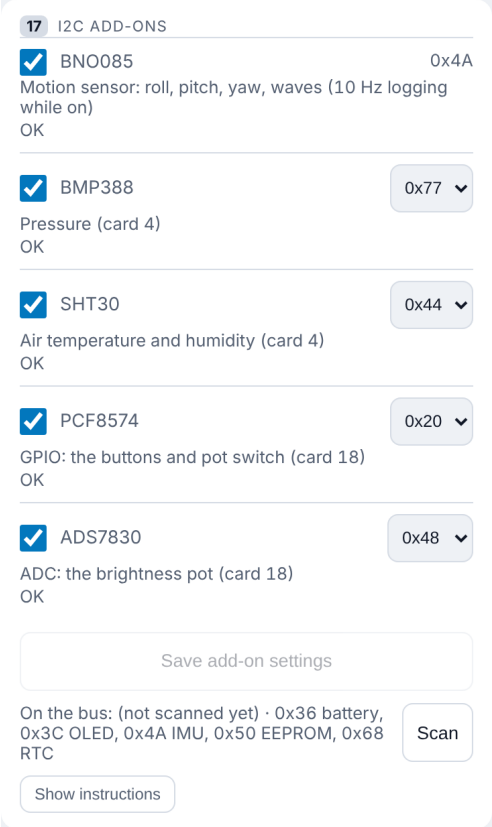
Where the VMR is: its distance from midships (fore or aft), from the centerline, and its height above the keel. This is used to move the heave to midships (card 6) to better estimate wave height and

Card 16 – Level Zero



Used to level the VMR when the vessel is in a static condition. Set zero button allows automatic setting with buttons for 0.1° changes

Card 17 – I2C Add-Ons

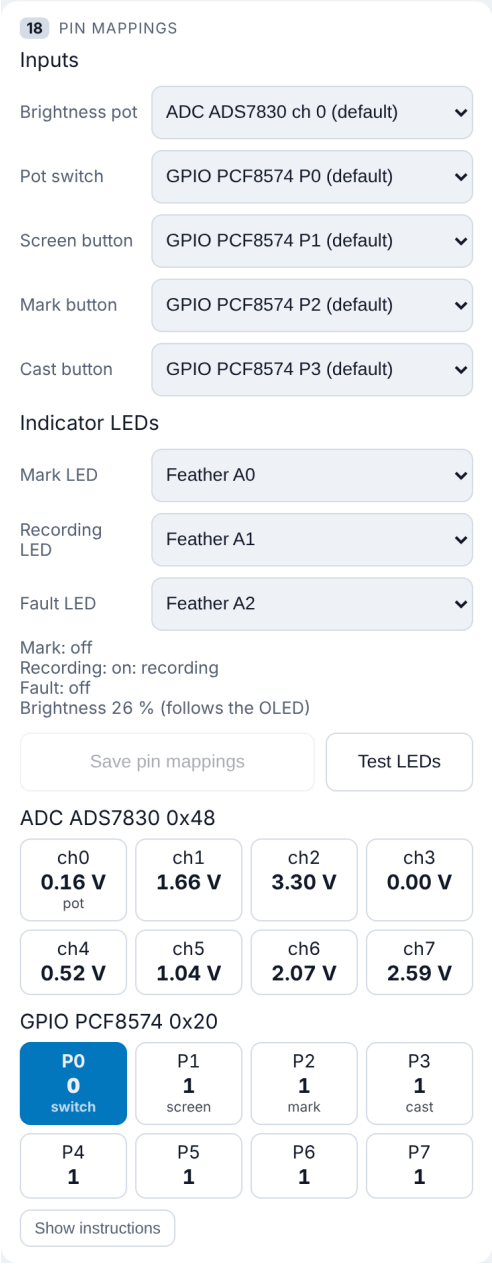


Select which I2C devices are in the system, or enable and disable boards

If an IMU is in the system, but the user doesn't want IMU data and to log slower than 10 Hz the IMU can be disabled thus allowing longer datalogging rates

Allows any I2C sensor to be added

Card 18 – Pin Mappings



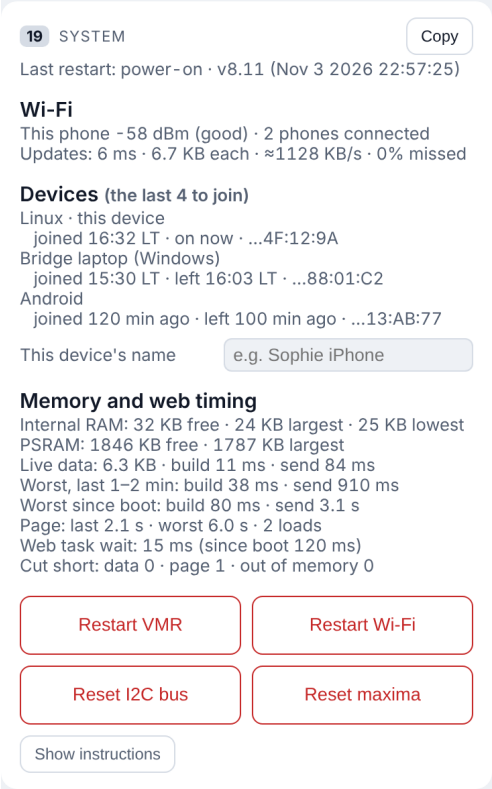
Which pins the controls and indicator LEDs use: the brightness pot (ADC), the pot switch, screen, Mark and Cast buttons (GPIO), and the Mark, Recording and Fault LEDs (Feather pins A0–A3)

Shows what each LED is showing and why the Fault LED is on, with the ADC channels and GPIO pins live. Test LEDs lights each LED in turn for 1 s

If built as default, no change in settings is required

Allows easy expandability for other sensors, controls, or indicators

Card 19 – System

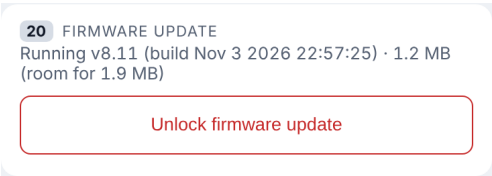


System-level information about the connected device and other connected devices

Memory diagnostics for help with troubleshooting

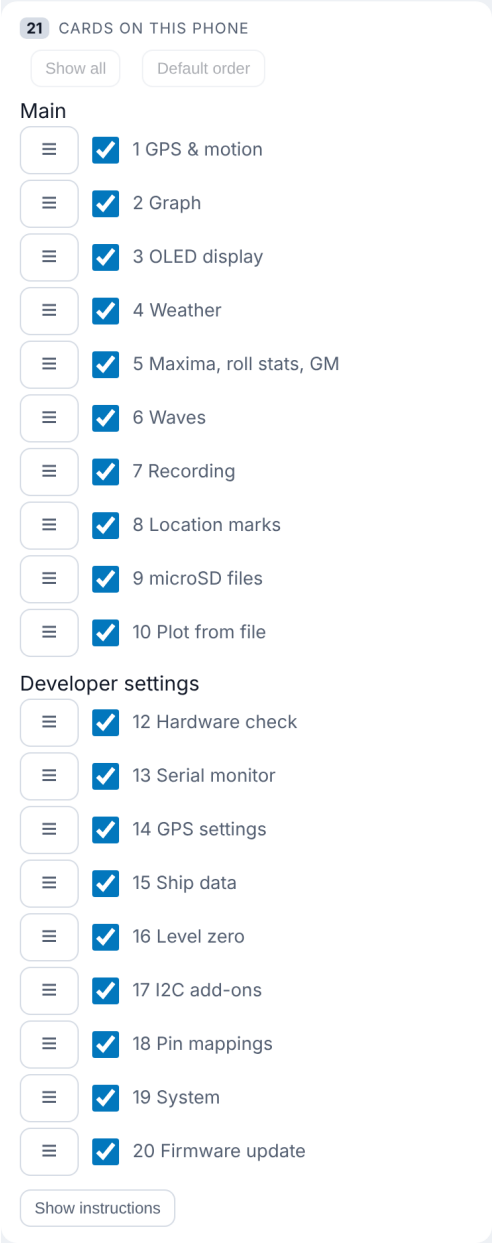
Buttons to control the systems

Card 20 – Firmware Update



Upload compiled .bin files to reflash the firmware over the air via Wi-Fi. Unlock firmware update asks for the firmware password first.

Card 21 – Cards on This Phone

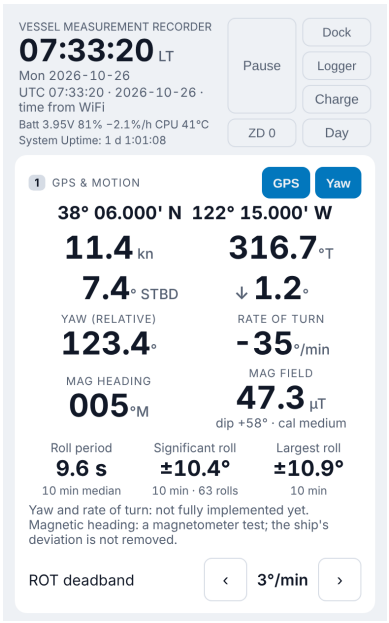


Untick a card to hide it on the connected device. Drag it by its ≡ handle to change the order. Each connected device keeps its own choice (does not change the order of cards for any other connected devices)

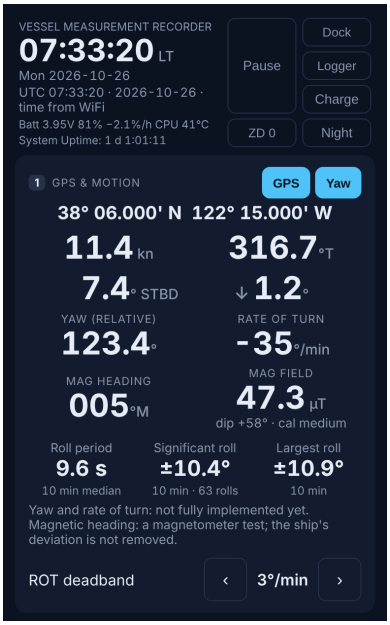
Hiding a card only hides it: recording, marks, the OLED and the LEDs carry on.

Day / Night / Red Mode

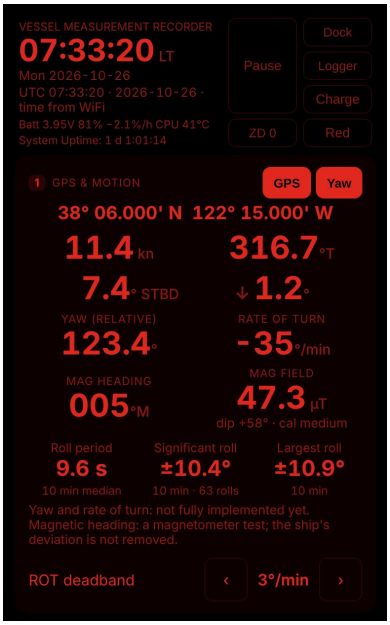
Changes the colors for everything – even the generated plots auto-change based upon the setting at the top



Day



Night

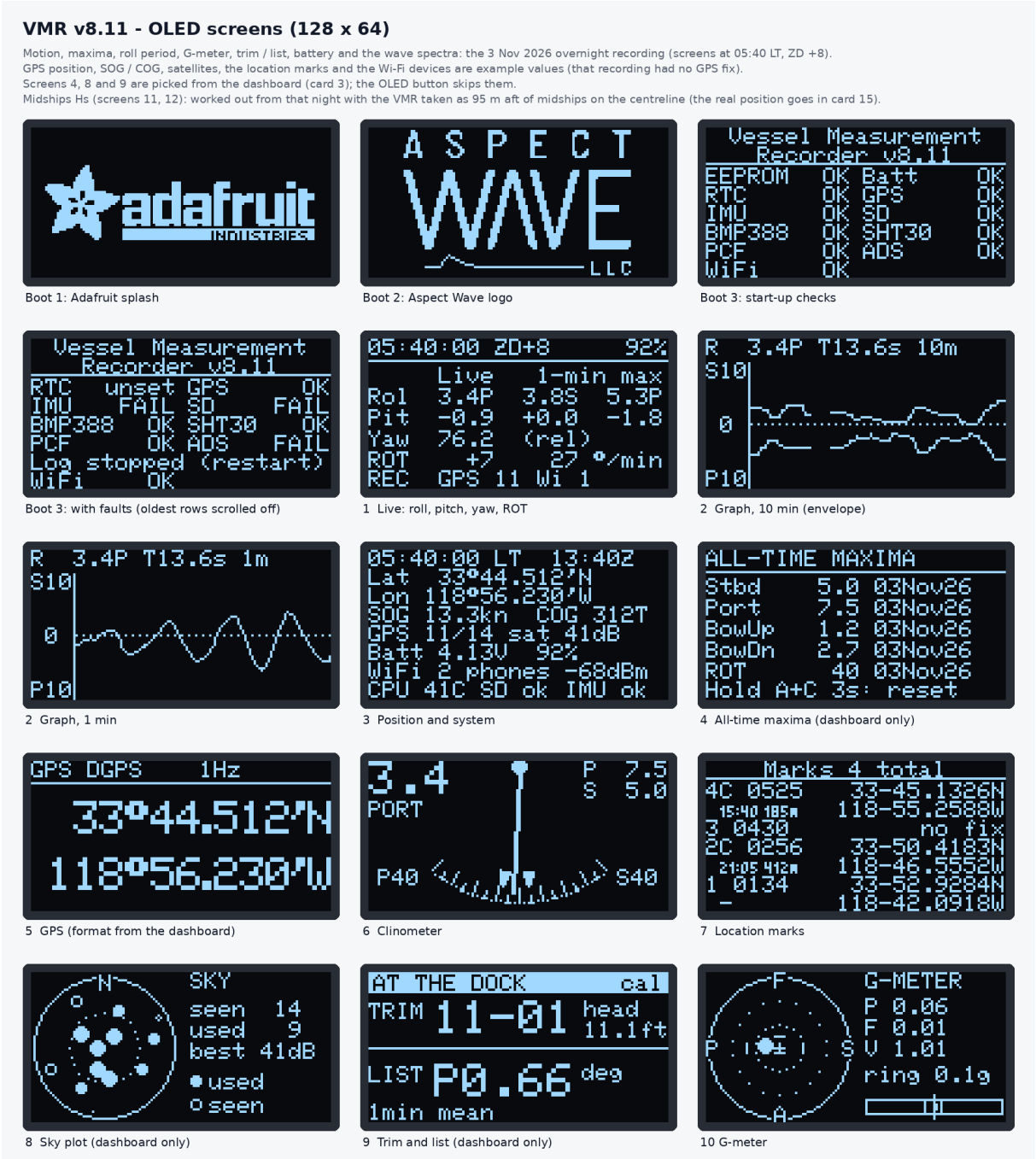


Red

OLED Screens

The possible screens that can be displayed on the OLED

Top three are during boot



Boot 1: Adafruit splash

Boot 2: Aspect Wave logo

Boot 3: start-up checks

Boot 3: with faults (oldest rows scrolled off)

1 Live: roll, pitch, yaw, ROT

2 Graph, 10 min (envelope)

2 Graph, 1 min

3 Position and system

4 All-time maxima (dashboard only)

5 GPS (format from the dashboard)

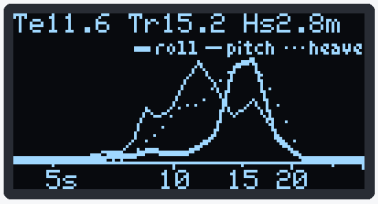
6 Clinometer

7 Location marks

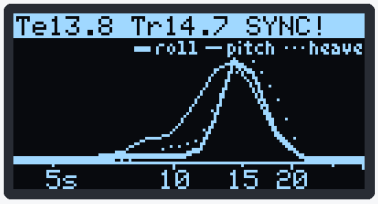
8 Sky plot (dashboard only)

9 Trim and list (dashboard only)

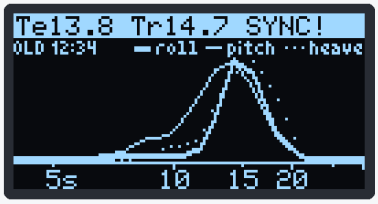
10 G-meter



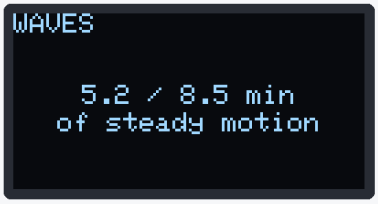
11 Waves (10:29 UTC), Hs at midships



11 Waves: Te near Tr, flagged (09:27 UTC)



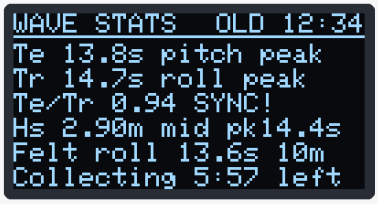
11 Waves: last estimate kept, 12:34 old



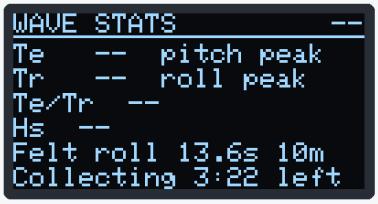
11 Waves: collecting, none yet



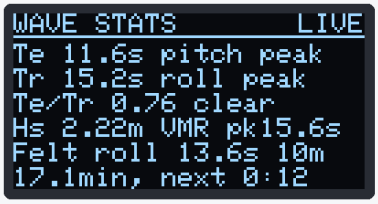
12 Wave stats (10:29 UTC), Hs at midships



12 Wave stats: last estimate, refilling



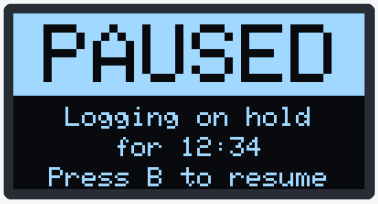
12 Wave stats: collecting, none yet



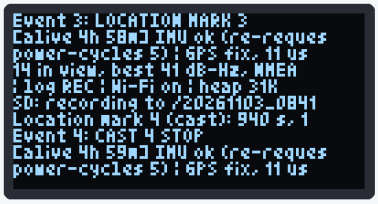
12 Wave stats: VMR position not entered



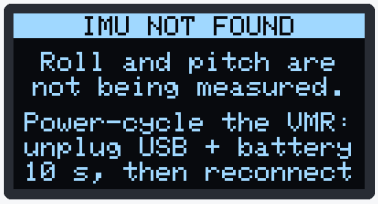
13 Wi-Fi devices (example devices)



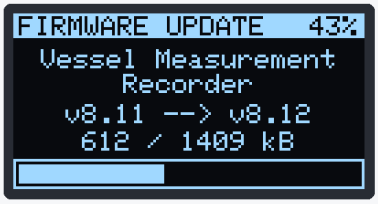
Paused (over every screen)



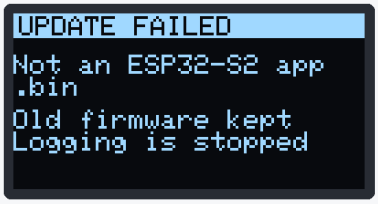
Serial monitor on the OLED



IMU not found (boot)



Firmware update



Firmware update failed



Datalogger: screen button, 5-min rows



Datalogger: screen button, 10 Hz

Boards and I2C addresses

Board	Connection	Address	Notes
ESP32-S2 Feather	–	0x36	MAX17048 battery gauge on the board
Adalogger FeatherWing	Feather headers	0x68	Real-time clock (CR1220); microSD chip select pin 10
Ultimate GPS FeatherWing	Feather headers	–	Serial1 (RX / TX), 9600 baud; active antenna on the u.FL
OLED FeatherWing 128 × 64	Feather headers	0x3C	Buttons A = pin 9, B = pin 6, C = pin 5
24LC32 EEPROM	STEMMA QT	0x50	Settings and maxima
BNO085	STEMMA QT	0x4A	Motion sensor
PCF8574 (GPIO)	STEMMA QT	0x20 [0x20–0x27]	Buttons and pot switch
ADS7830 (ADC)	STEMMA QT	0x48 [0x49, 0x4B]	Brightness pot. Not 0x4A: that is the BNO085
BMP388 (optional)	STEMMA QT	0x77 [0x76]	Pressure
SHT30 (optional)	STEMMA QT	0x44 [0x45]	Air temperature and humidity

Addresses in brackets are set with the jumpers on each board; pick the same address in card 17. Each board must have its own address; the defaults all work together.

Feather pins

Pin	Connection
A0 (GPIO 18) A1 (17) A2 (16)	Mark LED Recording LED Fault LED
A3 (GPIO 15)	Free; can be an LED (card 18) or an available analog input
A4 (GPIO 14)	PCF8574 INT (wake-up)
A5 (GPIO 8)	Pot wiper, if wired to the Feather, otherwise an available analog input
5, 6, 9	OLED buttons C, B, A
10	microSD chip select
11, 12	Pot switch and screen button, if wired to the Feather
RX / TX	GPS (through the stack)
SDA / SCL	I2C: STEMMA QT port and the stack

Wiring

Item	Connect	Notes
Pot switch	PCF8574 P0 and GND	Closed = OLED on; open = OLED and LEDs off
Screen button	PCF8574 P1 and GND	Next OLED screen
Mark button	PCF8574 P2 and GND	One mark per press
Cast button	PCF8574 P3 and GND	Hold Cast and press Mark to start or stop a cast
PCF8574 INT	Feather A4	Lets the buttons wake the VMR from Datalogger sleep
Brightness pot (10 kΩ linear)	Wiper to ADS7830 channel 0; ends to VIN and GND	OLED and LED brightness
Mark LED	Feather A0 → 1–2.2 kΩ → LED → GND	
Recording LED	Feather A1 → 1–2.2 kΩ → LED → GND	
Fault LED	Feather A2 → 1–2.2 kΩ → LED → GND	

These are the defaults; card 18 changes them, so write down what you used.

What Header to Solder each Board

Photos

Terminal blocks to I2C breakouts and OLED Featherwing

- 8-pin block on the PCF8574's P0–P7 pads.
- 8-pin block on the ADS7830's A0–A7 pads.

4-pin block on the Qwiic breakout.

Adalogger

- Fit the CR1220 coin cell (+ side up). This keeps the clock running when the VMR is off.
- Insert the microSD card.

[Photo: Adalogger with coin cell and microSD card]

-

[Photo: PCF8574 terminal block]

[Photo: ADS7830 terminal block]

Assembly

Enclosure

1. Plan the layout so the OLED is visible through the lid and the BNO085 is mounted flat and square to the box
2. Drill holes for the knob, buttons, power switch, LEDs, USB-C panel mount, both antennas, and PG-7 cable gland

3. Route the GPS antenna cable and Wi-Fi antenna cable and tighten. Connect the external Wi-Fi antenna (w.FL) on the Feather and the active GPS antenna (u.FL) on the GPS FeatherWing
4. Route the SHT30 cable through the PG-7 cable gland and tighten
5. Stack the Adalogger, GPS with CR1220 battery, and OLED FeatherWing on the ESP32-S2 Feather, checking every pin lines up before pushing down. The stack shares power, I2C and the GPS serial line through the headers, so no wires are needed between them
6. Chain the STEMMA QT boards from the OLED's STEMMA QT port, in any order: EEPROM, BNO085, PCF8574 and ADS7830, then the optional BMP388 and SHT30. Arrange them to suit the enclosure. The Feather's STEMMA QT port is switched on by the firmware at start-up; that is normal.
7. Fit and solder the latching power switch between the battery and the Feather
8. Mount the BNO085 flat, chip side up, with its X arrow toward the bow (see Mounting).

[Photo: full wiring diagram]

[Photo: stack assembly]

[Photo: STEMMA QT chain]

Buttons and pot switch

Each button and the pot switch connect between a PCF8574 pin and GND. Use an arcade quick-connect wire for each: push the quick-connect ends onto the switch's two terminals, then screw the other ends into the PCF8574's terminal block, one on the pin (see Wiring table above) and one on GND. No soldering or resistors are needed: the PCF8574 holds the pins high and a pressed button pulls them low.

Wake-up wire: PCF8574 INT to Feather A4. Without it the buttons still work, but they can't wake the VMR from Datalogger sleep (OLED button B always can).

[Photo: PCF8574 button wiring]

[Photo: INT to A4 wire]

Brightness knob

The knob is a potentiometer with an on/off switch on the same shaft. Pot ends to the ADS7830's VIN and GND, the wiper (middle pin) to channel 0. Swap the two ends to reverse which way is brighter.

Without the I2C boards, the controls can go on the Feather itself: screen button to pin 12 and GND, pot switch to pin 11 and GND, and the pot wiper to A5 with its ends on 3V and GND. A5 is used because A0–A4 are on the ESP32's second ADC, which does not read reliably while Wi-Fi runs (as LED outputs they are fine). Never connect the pot to BAT or USB: over 3.6 V on A5 can damage the ESP32-S2.

[Photo: pot wiring]

Indicator LEDs

Each LED has its own Feather pin rather than a PCF8574 pin, so the Fault LED still works when the I2C bus does not. LEDs with resistors toward 2.2 kΩ keep the system darker. A3 is free for another LED or analog input.

Power switch

Splice the latching switch into the battery's red (+) wire; the black (−) wire goes straight to the JST plug. Cut and solder one wire at a time so the red and black never touch.

- Switch off: the battery is fully disconnected and draws nothing.
- With the switch off, plugging in USB still runs the VMR, but the battery only charges with the switch on. Charge mode on the dashboard charges fastest.

[Photo: power switch in battery lead]

Covering board LEDs (optional)

The BNO085, EEPROM, and other I2C boards have green power LEDs that can't be turned off in software. To darken the system, cover them with black electrical tape or a dab of black nail polish.

[Photo: covered LEDs]

Mounting on a Vessel

The VMR measures the box's motion, so the BNO085 must be fixed rigidly to the enclosure, and the enclosure rigidly to the ship. Any wobble shows up as false roll and pitch

- BNO085 flat, chip side up, X arrow toward the bow (Y then points to port)
- Roll shows as S (starboard side down) or P (port side down); in the CSV, + is starboard down. Pitch + is bow up. Rate of turn + is turning to starboard
- Check once: starboard down reads S, bow up reads +, a turn to starboard reads ROT +. If one is backwards, set ROLL_FLIP, PITCH_FLIP or YAW_FLIP to 1 in the sketch or via the Wi-Fi dashboard's Developer Settings
- Enter where the VMR is in card 15: from midships, from the centerline and above the keel.
- With the vessel upright and still, set level zero in card 16 (it uses a 5-s mean of roll and pitch), or calibrate the trim from the draft marks at the dock (card 11)
- Place the GPS antenna with a clear view of the sky
- Location on the ship: on a fixed, rigid surface, e.g. the bridge near the centerline

[Photo: BNO085 mounting]

[Photo: VMR installed on the bridge]

Before you start: Arduino IDE

Needed only to build the firmware from the source code. A VMR that is already running can be updated over Wi-Fi instead (Firmware updates, below)

1. Install Arduino IDE 2.x from arduino.cc
2. Add the ESP32 boards: in File > Preferences (Arduino IDE > Settings on a Mac), paste `https://espressif.github.io/arduino-esp32/package_esp32_index.json` into "Additional boards manager URLs". Then in Tools > Board > Boards Manager, search "esp32" and install "esp32 by Espressif Systems". The VMR is built and tested with version 3.3.12
3. Install the following libraries in Tools > Manage Libraries, accepting the extra libraries when asked

Adafruit SH110X (brings Adafruit GFX and BusIO)	RTClib
Adafruit GPS Library	Adafruit BNO08x
Adafruit MAX1704X	Adafruit BMP3XX
Adafruit SHT31 (also drives the SHT30)	Adafruit PCF8574
Adafruit ADS7830 (with Adafruit Unified Sensor)	SD,

4. Under Tools: Board "Adafruit Feather ESP32-S2", USB CDC On Boot "Enabled", Upload Mode "Internal USB", Partition Scheme "Minimal SPIFFS (1.9MB APP with OTA/128KB SPIFFS)". The default partition scheme (TinyUF2 No OTA) runs but has no room for updates over Wi-Fi

5. Put VMR.ino in a folder called VMR (the folder and file names must match) and open it
6. Plug the Feather in with a USB-C cable that carries data (some are charge-only). Put it into upload mode: double tap RESET. Select the port under Tools > Port and click Upload. The first compile takes a few minutes
7. When the output stops moving and says, “Hash of data verified”, press RESET: the sketch won’t start until you do. The port changes; reselect it and open Tools > Serial Monitor at 115200 baud. The boot list shows OK or FAIL for each part; with only the bare Feather connected, FAIL for the rest is normal

[Photo: Arduino IDE board settings]

If uploads fail

- **“Failed to connect” or no port:** try double-tapping the RESET button so the drive re-connects to the computer, select the new port, upload.
- **“Port busy”:** close the Serial Monitor and anything else using the port, then try again.
- **The port keeps disappearing after RESET:** Unplug from all power for ~10 seconds then try again

Firmware updates

1. In the Arduino IDE: Sketch > Export Compiled Binary.
2. Join the VMR’s Wi-Fi and open the dashboard. In card 20, unlock with the password “vvvv”
3. Choose VMR.ino.bin from the sketch’s build folder (not the .merged, .bootloader or .partitions file) and upload. The VMR restarts, and an open dashboard reloads itself when it is back (if your device is configured to auto-connect to the VMR’s Wi-Fi

A failed update leaves the old firmware running. EEPROM values are kept between restarts and firmware updates

Without the Arduino IDE: a new Feather can be loaded from the single file VMR.ino.merged.bin (same build folder), written at address 0x0 with esptool or Adafruit’s WebSerial ESPTool page in Chrome. Put the Feather in upload mode first (BOOT + RESET).

First start

1. With the knob switch on, depress the power switch,. The OLED shows the Adafruit and Aspect Wave logos, then the start-up checks: each part OK or FAIL. Any FAIL holds the list for 12 s so you can read it.
2. Join the Wi-Fi network “Vessel Measurement Recorder” using password “somestupidcode”
3. Open http://192.168.4.1. The hotspot allows up to 4 connected devices
4. Open Developer Settings. In card 17, press Scan to list every address that answers, tick the parts fitted, check their addresses and Save. The Weather card appears once the BMP388 or SHT30 is on.
5. In card 18, check the pins match the wiring, then Test LEDs.
6. Test each control: the knob changes the OLED brightness and its switch turns the OLED plus LEDs off and on; the screen button changes screens; Mark shows MARK; hold Cast and press Mark for CAST START, and again for CAST STOP.
7. Take the VMR outside or near a window and wait for a GPS fix; the clock sets itself from the GPS.
8. In card 15, enter the ship data and where the VMR is (the keel-to-bridge height and mean draft give the sea-level pressure).
9. Set level zero in card 16, or calibrate at the dock in card 11.

10. Start a short recording in card 7 and check the file in card 9.

How the numbers are worked out

- **Waves:** every 30 s from samples twice a second over the last 17 min (the first after 8.5 min of steady motion; again after a pause or IMU gap; off at the dock). Te is the peak of the pitch spectrum, the pitching period (IMO MSC.1/Circ.1228, 1.6). Tr is the peak of the roll spectrum, the natural roll. The roll period in card 1 times rolls one by one and includes rolling forced at the wave period, so it reads shorter. Te / Tr is flagged within ±15 % of 1 or of ½ (Circ.1228 4.2.3 gives no tolerance)
- **Heave Hs:** 4 × the standard deviation of heave from the vertical acceleration, periods 3–20 s. With the VMR's position in card 15 it is moved to midships, taking off the rise and fall that pitch and roll give a VMR away from midships. A long hull hardly heaves in waves much shorter than itself, so Hs is a guide, not the wave height; long swell comes closest
- **GM:** the IMO rolling-period formula (IS Code 2008, A 2.3): $GM = (2 \cdot C \cdot B / T)^2$, $C = 0.373 + 0.023 \cdot B/d - 0.043 \cdot L/100$, with B the beam, d the mean draft, L the LBP (LOA if none) and T the roll spectrum peak. A cross-check only
- **Ship data checks:** the Cb the displacement implies, $\Delta = Cb \times L \times B \times d \times 1.025$ t/m³ (the quarter-mean draft when a mid draft is given), pointed out when more than 0.06 from the Cb entered; the Cwp the TPI implies, $TPI \times 39.03$ m² / (L × B)
- **Roll statistics:** each full roll counts once; its amplitude is half its peak-to-peak swing about the running mean; the significant roll is the mean of the highest third
- **Trim at the dock:** L × tan(pitch), averaged over the window. Calibrating stores the pitch offset that makes the reading match the drafts; 0.1° is about 1.6 ft on 280 m
- **Weather:** 2-s means kept for 20 min and 30-s means for 3 h. Sea-level pressure adds the sensor's height above the waterline (keel to bridge minus the mean draft). The 3-h change uses a sample every 10 min and counts as steady within ±0.5 hPa. Dew point by the Magnus formula
- **Magnetic heading:** tilt-corrected with the IMU's roll and pitch; the ship's deviation is not removed (still being tested)

Recordings and files

- **Names:** /YYYYMMDD_HHMMSSZ_name.csv, from the UTC time Start was tapped (NOCLK_001… without a clock). A recording cut off by a power loss is not resumed
- **Header:** lines starting with # hold the firmware, ship data, calibration and notes on the columns. Read a file with `pd.read_csv(path, comment='#')`
- **Rows:** GPS columns are filled only on the row after a new fix, and the wave columns only on the row after each new estimate, so nothing is duplicated
- **Card outages:** rows wait in a 256 KB memory buffer (about 3 min at 10 Hz) and are written when the card is back. The first row after is an event "SD RECOVERED after N s (M rows lost)"
- **Location marks:** folder location_mark, one CSV per list, named by the first mark's local time. Local time, ZD and UTC, the position also in decimal degrees, satellites, HDOP, fix age and weather data (if sensors are installed and selected)
- **Downloads:** several files download as one .zip. One day at 10 Hz is about 80 MB; for weeks of data, it's recommended to remove the card from the system

Troubleshooting

Problem	What to do
A part shows FAIL at start-up	Check its STEMMA QT cable is fully seated in at both ends. Scan in card 17 shows which addresses answer. Missing parts are retried every 5 s, so a reseated cable can recover without a restart.
Roll and pitch freeze	The VMR asks the IMU again, then power-cycles the I2C boards by itself; Reset I2C bus in card 19 does it by hand. If the IMU never starts, unplug USB and battery for 10 s
No GPS fix	Check the antenna adapter is seated on the GPS FeatherWing and the antenna can see the sky. The first fix can sometimes take several minutes
GPS antenna shows internal	Only an active antenna shows as external: check the cable is fully-seated. The GPS board's red LED blinks every 1 s while searching, every 15 s with a fix
Mark button doesn't wake it from Datalogger mode	Check the INT-to-A4 wire. OLED button B always wakes it
A phone can't join the Wi-Fi	The hotspot allows 4 connected devices. Restart Wi-Fi or the system in card 19
Dashboard won't load or acts old	Make sure the phone is on the VMR's Wi-Fi (not auto-connecting to another network) then reload the page
Recording LED blinks fast	The microSD card is failing; rows are kept in memory for about 3 min. Check or replace the card
Dashboard slow	Card 19 memory and web timing: slow Updates with short send times mean a weak Wi-Fi signal. Move closer
Can't upload over USB	See If uploads fail, above
Firmware update failed	The old firmware keeps running. Try installing again with the phone closer to the VMR, leaving the Wi-Fi dashboard open. Try the VMR.ino.bin, the app image

Equations Used by Firmware

The following are all the equations used by the firmware in calculating values. Angles are in degrees unless marked; lengths in meters.

Sign Conventions

With the BNO085 flat, chip side up and its X arrow toward the bow (Y to port, Z up):

Quantity	Positive
φ roll	Starboard side down (shown S); negative = port down (P)
θ pitch	Bow up
Yaw	Clockwise from above, like a heading. Relative (gyro and accelerometer, no magnetometer), so it drifts slowly
Rate of turn	Turning to starboard, °/min
acc_fwd, acc_stbd, acc_up	Felt force (gravity included) in ship axes, g. Level and still = 0, 0, +1

If one reads backwards, set ROLL_FLIP, PITCH_FLIP or YAW_FLIP to 1 in the sketch.

Level Zero

Roll and pitch run through a 5-s running mean. Set level zero stores that mean as the offset, so

$$\varphi_{shown} = \varphi_{raw} - \varphi_0 \qquad \theta_{shown} = \theta_{raw} - \theta_0$$

The card 16 arrows nudge the offsets 0.1° per tap. Use it only with the vessel upright and still: any list or trim at that moment is removed from every reading.

Roll Period (Card 1)

The roll period as felt, timed one roll at a time:

- Roll is low-passed by two first-order stages (τ = 0.15 s each) to remove vibration.
- A mean that follows a slowly changing list without lagging is removed (60 s):

$$m_1 \leftarrow m_1 + k(r - m_1) \qquad m_2 \leftarrow m_2 + k(m_1 - m_2) \qquad k = \frac{\Delta t}{60 + \Delta t}$$

$$x = r - (2m_1 - m_2)$$

- A roll is the time between upward zero-crossings of x, interpolated between samples. It counts only if x swung past ±0.3° both ways and lasted 2–60 s.
- The roll period is the median of the last 9 rolls (at least 3). It clears after 60 s without a roll or after a data gap.

Rolls timed one by one include rolling forced at the wave encounter period, so this reads shorter than the natural roll. GM and the Te / Tr check use the roll spectrum peak instead.

Roll Statistics (Card 5)

Over the last 10 or 30 min, each roll counted above gives one amplitude, half its peak-to-peak swing about the running mean:

$$A_i = \frac{x_{max} - x_{min}}{2}$$

The significant roll is the mean of the highest third, like significant wave height:

$$\varphi_{1/3} = \frac{1}{k}\sum_{i=1}^{k} A_{(i)} \qquad k = \lceil n/3 \rceil$$

with the amplitudes sorted largest first. The mean list and RMS roll come from 10-s sums of roll and roll²:

$$\varphi_{mean} = \frac{S_1}{N} \qquad \varphi_{RMS} = \sqrt{\frac{S_2}{N} - {\varphi_{mean}}^2}$$

Draft

One draft is used for GM, sea-level pressure, freeboard, air draft and the hull-form checks: the quarter mean when all three drafts are entered (v8.12), as a sagging hull displaces more than the mean of the ends says:

$$d = \frac{F + 6M + A}{8} \text{ with a midships draft;} \qquad \text{otherwise } d = \frac{F + A}{2}$$

where

Symbol	Meaning
F, M, A	Drafts forward, midships and aft read at the marks
d	The draft used (m)

GM Estimate (Card 5)

The IMO rolling-period formula (IS Code 2008, Part A, 2.3):

$$T = \frac{2CB}{\sqrt{GM}} \quad \Rightarrow \quad GM = \left(\frac{2CB}{T}\right)^2$$

$$C = 0.373 + 0.023\frac{B}{d} - 0.043\frac{L}{100}$$

where

Symbol	Meaning
B	Beam (m), card 15
d	Draft (m), as above
L	LBP (m), or LOA if there is no LBP
T	Natural roll period (s): the peak of the roll spectrum (Waves), not the card 1 roll period
GM	Metacentric height (m); shown in feet (× 3.2808) with metres beside it

No GM without the beam, a draft and a length; for the first 8.5 min of steady motion; at the dock, paused or with the IMU off; or from an old wave estimate. With the roll peak within ±15 % of Te, GM is still given, noted “Te ≈ Tr: near resonance”.

Why the spectrum peak: on 1 November 2026 (loaded) the card 1 roll period was 13.7 s against a 14.6 s spectrum peak, giving 18–19 ft and 15.4 ft against CargoMax’s 14.48 ft. Overnight on 3 November the mean was 14.14 ft. A cross-check only, not a stability calculation.

Wave Analysis (Card 6)

Every 0.5 s the VMR keeps 0.5-s means of the vertical acceleration in the earth frame, pitch and roll. The acceleration is the accelerometer turned by the orientation quaternion, less gravity:

$$a_z = (R(q)\,a)_z - g, \quad \text{g = 9.80665 m/s²}$$

The window holds the last 2048 samples (17.07 min). A gap, a pause, an IMU dropout or Dock mode starts it again. From 1024 samples (8.5 min) on, Welch spectra are worked out every 30 s: segments of N = 256 samples (128 s), half overlapped, each with its mean removed and a Hann window, then an FFT:

$$w_i = 0.5 - 0.5\,cos\frac{2\pi i}{N}$$

$$S(f_k) = \frac{2|X_k|^2}{f_s \sum {w_i}^2} \cdot \frac{1}{n_{seg}} \qquad \Delta f = \frac{f_s}{N} = \frac{1}{128}\,\text{Hz}$$

with fs = 2 Hz. Bins k = 4 to 80 are used: periods of 32 s down to 3.2 s. The peak of each spectrum is refined with a parabola through the largest bin (b) and its neighbours (a, c):

$$\delta = \frac{a-c}{2(a-2b+c)} \qquad T_p = \frac{1}{(k_b+\delta)\Delta f}$$

Period	From
T_e	Peak of the pitch spectrum: the pitching period, used as the wave encounter period (IMO MSC.1/Circ.1228, 1.6)
T_r	Peak of the roll spectrum: the natural roll period
T_z	Peak of the heave spectrum

Synchronous and Parametric Rolling

SYNC! if $\left|\frac{T_e}{T_r} - 1\right| \le 0.15$ PARAM! if $\left|\frac{T_e}{0.5T_r} - 1\right| \le 0.15$

MSC.1/Circ.1228, 4.2.3 names these conditions but gives no tolerance; ±15 % is the VMR's choice.

Heave Hs at the VMR

The heave spectrum comes from the acceleration spectrum by dividing by (2πf)⁴. Below 0.05 Hz (periods over 20 s) it is set to zero, as integrating there only amplifies noise:

$$S_z(f) = \frac{S_a(f)}{(2\pi f)^4} \qquad m_0 = \sum_k S_z(f_k)\,\Delta f \qquad H_s = 4\sqrt{m_0}$$

Heave Hs at Midships

With the VMR's position entered in card 15, the hull is taken as rigid. For small angles the VMR rises and falls with the heave at midships plus what pitch and roll give it:

$$z_{VMR} = z_{mid} + x\theta - y\varphi$$

So, frequency by frequency, from each segment's complex spectra (which keep the phase between the three):

$$Z_{mid}(f) = \frac{A(f)}{-(2\pi f)^2} - x\Theta(f) + y\Phi(f)$$

where

Symbol	Meaning
x	VMR distance from midships (m), + forward
y	VMR distance from the centreline (m), + starboard
A, θ, Φ	Complex spectra of vertical acceleration, pitch and roll (pitch and roll in radians)

|Zmid|² is averaged over the segments and scaled like the other spectra, then Hs at midships = 4√m0 as above. Pitch and roll are averaged over the same 0.5 s as the acceleration so they line up in time. Card 6 shows Hs at midships when the position is entered, otherwise at the VMR. The height above the keel is stored but not used here.

A long hull hardly heaves in waves much shorter than itself, so Hs is a guide to the sea, not the wave height; long swell comes closest. On 3 November, with x = −95 m, Hs at midships read about 0.1–0.7 m higher than at the VMR.

Ship Data Checks (Card 15)

$$V = L\,B\,d \qquad C_b = \frac{1.01605\,\Delta_{LT}}{1.025\,V} \qquad \Delta_t = 1.025\,C_b\,V$$

$$C_{wp} = \frac{39.03\,\cdot\,TPI}{L\,B}$$

$$\text{hog / sag} = M - \frac{F+A}{2} \qquad \text{freeboard} = D_{deck} - d \qquad \text{air draft} = H_{top} - d$$

where

Symbol	Meaning
V	Box volume at the draft (m³)
Δ_{LT}, Δ_t	Displacement in long tons and in tonnes (1 LT = 1.01605 t; sea water 1.025 t/m³)
TPI	Long tons per inch immersion; 1 LT/in in sea water = 39.03 m² of waterplane
$hog\ /\ sag$	+ = sagging
D_{deck}, H_{top}	Keel to main deck, keel to the very top

Saving points out: a Cb from the displacement outside 0.45–0.92 (is it displacement, not deadweight, and LT / MT right?), or more than 0.06 from the Cb entered; a Cwp from the TPI outside 0.6–1.0; a trim over 3 % of L or a hog / sag over L/250; keel heights out of order.

Trim and List at the Dock (Card 11)

Pitch and roll are averaged into 1-s means, then over the chosen window (30 s to 5 min):

$$t = L\,\tan(\theta - \theta_0) \qquad \text{list} = \varphi_{mean} \qquad \text{across the beam} = B\,\tan|\,\varphi_{mean}\,|$$

Trim t is + by the stern, with L the LBP (or LOA). Calibrating from the drafts read at the marks (F, A in inches) stores the offset that makes the window's mean pitch θmean match them:

$$\theta_{true} = \tan^{-1}\frac{0.0254\,(A-F)}{L} \qquad \theta_0 = \theta_{mean} - \theta_{true}$$

Without a calibration the trim follows level zero. 0.1° is L tan 0.1° = 0.49 m, about 1.6 ft on 280 m. The sensor reads the deck slope where it is mounted and hull bending changes that slope near the ends, so recalibrate when the loading changes much.

Weather (Card 4)

Sea-level pressure, by the barometric formula with the standard lapse rate:

$$P_0 = P\left(1 - \frac{0.0065\,h}{T + 0.0065\,h + 273.15}\right)^{-5.257} \qquad h = H_{bridge} - d$$

where

Symbol	Meaning
P	Pressure at the BMP388 (hPa); the CSV keeps this, not the sea-level value
h	Sensor height above the waterline (m): keel to bridge less the draft
T	Air temperature (°C): the SHT30, else the BMP388's own sensor, else 15 °C

Dew point, by the Magnus formula (a = 17.62, b = 243.12 °C), with RH in %:

$$\gamma = ln\frac{RH}{100} + \frac{a\,T}{b + T} \qquad T_d = \frac{b\gamma}{a - \gamma}$$

Where the air temperature meets the dew point, expect fog or condensation. The 3-hour change uses a sample every 10 min:

$$\Delta P_{3h} = P_{now} - P_{-3h} \qquad \text{steady if } |\,\Delta P_{3h}\,| < 0.5 \text{ hPa}$$

Graphs keep 2-s means for 20 min and 30-s means for 3 h.

Magnetic Heading Test (Card 1)

The BNO085's calibrated field m (µT) is read at 10 Hz and turned into the earth frame with the orientation quaternion, which removes roll and pitch. The bow (the sensor's X axis) is turned the same way, b. The horizontal field is then split along the bow and to starboard:

$$w = R(q)\,m \qquad fwd = w_x b_x + w_y b_y \qquad stb = w_x b_y - w_y b_x$$

$$\psi_m = atan2(-stb, fwd) \qquad dip = atan2\left(-w_z, \sqrt{{w_x}^2 + {w_y}^2}\right)$$

fwd and stb are smoothed over about 2 s (the components, not the angle, which wraps at 360°). The ship's deviation is not removed, and on a steel hull it can be large: a test only. It does not feed roll, pitch or yaw.

Rate of Turn, G-Meter, SOG and COG

- **Rate of turn:** the gyro's Z rate × 60 (°/min), + to starboard. The screens and maxima smooth it over 1 s and show values inside the deadband as 0; the CSV keeps the raw rate.
- **G-meter:** the felt push in the deck plane (up = forward, right = starboard; gravity included, so a starboard heel puts the dot to starboard), smoothed over 0.3 s, with peak-hold ticks. The bar shows vertical g; 1 g = still.
- **SOG and COG:** from 1-Hz GPS over the averaging time (1–60 s, default 10 s). SOG is the mean speed; COG is the direction of the mean velocity, so it wraps through 000/360 properly and near-zero speeds carry little weight:

$$COG = atan2\left(\sum v\,\sin c\,, \sum v\,\cos c\right)$$

Indicator LED Brightness and Morse

5 kHz PWM, 10 bits. The duty follows the OLED brightness c (1–255, from the knob or the dashboard slider); the square evens out the steps to the eye and 12 keeps the LEDs just visible at the bottom:

$$duty = 12 + 1011\left(\frac{c}{255}\right)^2 \quad \text{(of 1023)}$$

FAULT in Morse is ..-. .- ..- .-.. - at 120 ms a unit (dot 1, dash 3, gaps 1 in a letter, 3 between letters, 7 between words): 52 units, about 6.2 s.

Constants

Constant	Value
Gravity g	9.80665 m/s²
Sea water density	1.025 t/m³
Long ton	1.01605 t
1 LT/in, sea water	39.03 m² of waterplane
Units	1 m = 3.2808 ft; 1 in = 0.0254 m; 1 hPa = 0.02953 inHg
Lapse rate; exponent	0.0065 K/m; 5.257
Magnus a, b	17.62; 243.12 °C
GM coefficient	C = 0.373 + 0.023 B/d − 0.043 L/100
Wave sampling	2 Hz; 2048-sample window (17.07 min); first after 1024 (8.5 min); every 30 s
Welch	256-sample segments (128 s), 50 % overlap, Hann window
Spectrum band	3.2–32 s (bins 4–80); heave cut below 0.05 Hz
Te / Tr tolerance	±15 %
Roll period	±0.3°, 60 s mean, median of 9 rolls; level zero 5-s mean